

# Peter Juhasz — Researcher

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📄 ResearchGate • in LinkedIn • 🐙 GitHub

## Summary

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I am a PhD Candidate in Mathematics at Aarhus University, supervised by Christian Hirsch. My doctoral research combines stochastic processes and topological data analysis to create novel models for complex networks. Prior to my PhD, I worked as a deep learning researcher in automated driving at Bosch Group Hungary, where I developed algorithms for vehicle perception, prediction and decision-making. Additionally, I led the design and implementation of scalable software solutions as a software architect in the same field.

## Education

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### PhD in Mathematics

*Aarhus University, Denmark, Stochastic Point Processes and Topological Data Analysis* 2022–

Supervisors: Dr. Christian Hirsch, Dr. Claudia Strauch

Thesis: Point Process Based Models of Evolving Higher-Order Networks

### Master of Business Administration with Highest Honors

*Budapest University of Technology and Economics, Hungary, Specialized in Finance* 2019–2021

Supervisor: Dr. Roland Molontay

Thesis: Prediction of Popularity of Memes using Machine Learning Methods

### Master of Science in Physics

*Budapest University of Technology and Economics, Hungary, Specialized in Applied Physics* 2013–2016

Supervisor: Dr. Gabor Vattay

Thesis: Examination of Spatially Embedded Complex Networks

Erasmus+ Scholarship: Delft University of Technology (Applied Physics)

### Bachelor of Science in Physics

*Budapest University of Technology and Economics, Hungary, Specialized in Applied Physics* 2010–2013

Supervisor: Dr. Szabolcs Beleznai

Thesis: Finite Element Method Plasma Simulation of Nitrogen Contaminated Ceramic Metal Halide Lamps

## Professional Experience

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### Research Assistant in Spatial Stochastics and Topological Data Analysis

*Department of Mathematics, Aarhus University, Denmark* 2022–

- Published 4 papers in stochastic models for higher-order networks
- Implemented topological data analysis algorithms on simulated spatial point processes
- Spent 3 months at TU Braunschweig supervised by Dr. Benedikt Jahnel (Weierstrass Institute for Applied Analysis and Stochastics, Berlin)
- Taught 4 courses in mathematics and computer science

### Deep Learning Researcher in Automated Driving

*Department of Automated Driving, Bosch Group Hungary* 2021–2022

- Patented a Gaussian regression based trajectory prediction model for automated driving  
Analyzed more than 20 000 hours of measurements of passenger car driving  
Enhanced prediction accuracy by a factor of 3 and the uncertainty estimation by a factor of 10
- Object detection and classification with ultrasonic sensors:  
Designed several features to represent objects and free space  
Implemented a U-Net neural network to detect and classify objects
- Real time object tracking with temporal convolutional neural networks:  
Improved a feature pyramid network to track objects in real time

## Software Architect in Automated Driving

Department of Automated Driving, Bosch Group Hungary

2018–2021

- Lead a team of 11 in environment prediction as a technical lead
- Created a concept for automated ground-truth generation for data-driven machine learning development
- Coordinated 2 teams in the integration of driver monitoring camera for hands-free driving

## Software Engineer for Mobile Networks

Business Unit IT & Cloud Products, Ericsson Telecommunications Hungary

2016–2018

- Analyzed response times of distributed database servers for optimal load balancing
- Coordinated a team of 8 as a deputy of the product owner
- Implemented performance critical algorithms for Smart Services Routers

## Publications, Patents

- C. Hirsch, B. Jahnel, S. K. Jhavar and **P. Juhasz**. *Poisson approximation of fixed-degree nodes in weighted random connection models*, Stochastic Process. Appl. (2025), 104593, DOI 10.1016/j.spa.2025.104593. 2025
- M. Brun, C. Hirsch, **P. Juhasz** and M. Otto. *Random connection hypergraphs*, arXiv:2407.16334, 2024, DOI 10.48550/arXiv.2407.16334. 2024
- K. Barnes, **P. Juhasz**, M. Nagy and R. Molontay. *Topicality boosts popularity: a comparative analysis of NYT articles and Reddit memes*, Soc. Netw. Anal. Min. **14** (2024), no. 119, DOI 10.1007/s13278-024-01272-3. 2024
- C. Hirsch and **P. Juhasz**, *On the topology of higher-order age-dependent random connection models*, arXiv:2309.11407, 2023, DOI 10.48550/arXiv.2309.11407. 2023
- P. Juhasz**, *Information propagation in stochastic networks*, Phys. A **577** (2021), 126070, DOI 10.1016/j.physa.2021.126070. 2021
- P. Juhasz**, *Method and device for predicting the trajectory of a traffic participant, and sensor system*, CNIPA: CN113988353A, EPO: EP3916697A1, JPO: JP2021190119A, USPTO: US2021366274A1, 2021. 2021
- P. Juhasz**, J. Steger, D. Kondor and G. Vattay, *A Bayesian approach to identify Bitcoin users*, PLoS One **13** (2018), e0207000, DOI 10.1371/journal.pone.0207000. 2018
- P. Juhasz**, Sz. Beleznaï and I. Maros, *Finite element method plasma simulation of nitrogen contaminated metal halide lamps*, Comsol Conference Proceedings, 2014. 2014

## Awards, Grants

### Individual Travel Grant

European Mathematical Society

2025

500 EUR – Talk "Random Connection Hypergraphs" at GPSD 2025 conference

### Academic Mobility Grant for International Exchange

Aarhus University

2024

20 000 DKK – Evolving Stochastic Network Models

### PhD Fellowship Grant (DDSA-PhD-2022-008)

Danish Data Science Academy

2022–2025

1 890 000 DKK – Topological Data Analysis Based Models of Evolving Higher-Order Networks

### Excellence Award for Outstanding Thesis and Diploma Work

Hungarian National Bank

2021

350 000 HUF – Digitalization, Artificial Intelligence, and the Age of Data – What Makes a Meme Viral?

## Conferences, Events

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### Talks

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|---------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Scaling Limits in Spatial Birth-Death Systems with Long-Range Interactions</b><br><i>Stochastic Interacting Particle Systems and Random Matrices</i> | 2025 |
| <b>Random Connection Hypergraphs</b><br><i>German Probability and Statistics Days 2025</i>                                                              | 2025 |
| <b>Stochastic Random Connection Models</b><br><i>Probability Seminar</i>                                                                                | 2024 |
| <b>Topological Data Analysis Based Models of Evolving Higher-Order Networks</b><br><i>Qualifying Exam</i>                                               | 2024 |
| <b>Adaptive Trajectory Prediction Based on a Gaussian Regression Model</b><br><i>Bosch Group: Bosch Innovation Day</i>                                  | 2020 |
| <b>Probabilistic Localization of Bitcoin Users</b><br><i>Hungarian Academy of Sciences: Statistical Physics Conference</i>                              | 2016 |

### Posters

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|------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Simplex Count Distribution in Random Connection Hypergraphs</b><br><i>Particle Systems in Random Environments</i>                           | 2024 |
| <b>Distribution of Betti Numbers in Age-Dependent Random Connection Models</b><br><i>DDSA Annual Conference</i>                                | 2024 |
| <b>Topological Data Analysis of Higher-Order Networks</b><br><i>Danish-Swedish Summer School on TDA and Spatial Statistics</i>                 | 2023 |
| <b>Dissecting Distributed Database Systems For Telecom Applications</b><br><i>Ericsson Telecommunications Hungary: Ericsson University Day</i> | 2016 |
| <b>Finite Element Method Plasma Simulation of Ceramic Metal Halide Lamps</b><br><i>Comsol Multiphysics Conference</i>                          | 2014 |

## Programming Skills

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|        |   |   |   |   |   |
|--------|---|---|---|---|---|
| Python | ● | ● | ● | ● | ● |
| Git    | ● | ● | ● | ● | ● |
| C++    | ● | ● | ● | ● | ● |

## Languages

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|           |   |   |   |   |   |
|-----------|---|---|---|---|---|
| Hungarian | ● | ● | ● | ● | ● |
| English   | ● | ● | ● | ● | ● |
| German    | ● | ● | ● | ● | ● |

## Teaching

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### **Advanced Probability Theory**

*Aarhus University, Course Code: F25.550171U020*

2025

10 ECTS Exercise Class, Spring Semester

### **Data Project**

*Aarhus University, Course Code: F24.550201U004*

2024

10 ECTS Computer Project Class, Spring Semester

### **Linear Transformations**

*Aarhus University, Course Code: E23.550171U010*

2023

10 ECTS Exercise Class, Autumn Semester

### **Ordinary Differential Equations**

*Aarhus University, Course Code: F23.550141U011*

2023

5 ECTS Exercise Class, Spring Semester

### **Advanced Calculus for Engineers**

*Aarhus University, Course Code: F23.280191U021*

2023

10 ECTS Exercise Class, Spring Semester